OIL SPILL DETECTION FROM SAR IMAGES BY DEEP LEARNING

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ABSTRACT

Oil spills, caused by accidents or by ships cleaning their tanks, represent big threats for maritime and coastal ecosystems health. A very effective detection of oil spills can be performed using satellite synthetic aperture radar (SAR) systems, operating regardless of cloud coverage and sunlight and capable of discriminating oil from regular sea surface. However, discriminating between real oil spills and look-alikes (such as natural oils and seepages, often occurring in upwelling sea areas), although well performed by expert SAR image interpreters, poses a great challenge for automatic processes. In addition, a visual check performed by human operators on a great number of images would be too expensive. Therefore, many solutions for automatic detection have been tried in the last few years, using probabilistic models and, more recently, machine learning. This work presents an innovative solution based on image-to-image translation using convolutional neural networks (CNNs) trained with an adversarial loss function. The proposed approach has been tested, with very promising results, using Radarsat-2 and Sentinel-1 SAR data over the Mediterranean Sea and some areas of the Atlantic Ocean and the North Sea.

Index Terms— SAR, Oil Spill, Deep Learning, Semantic Segmentation, U-Net, GAN

1. INTRODUCTION

During the last few decades, the amount of accidental or illegal discharges in the open sea have become a serious problem for maritime environment. Illegal actions are driven by the will to avoid the cost of a legal cleaning of ship's tanks, increasing the already massive pollution of the seas, causing threats to our health [1]. Satellite synthetic aperture radar (SAR) systems are very effective tools for oil spill detection. In fact, oil and regular sea surface can be distinguished in SAR images due to the very different backscattered signals. Moreover, SAR systems can operate almost regardless of cloud coverage and sunlight. However, discriminating in SAR images between oil spills and look-alikes (such as natural oils and seepages, often occurring in upwelling sea areas), although well performed by expert SAR data interpreter, pose a great challenge. Moreover, it is clearly too costly for human operators to visually examine a huge number of images.

Early automatic detection methods from satellite synthetic aperture radar (SAR) data were based on probabilistic models, involving the detection of dark spots in the image based on a threshold, followed by a classification as spill or look-alike depending on physical features [2]. More recent solutions exploited machine learning algorithms, such as Support Vector Machines (SVM), for final classification; however, a manual extraction of the features was still needed [3], a tedious and error-prone operation.

Following the diffusion of convolutional neural networks (CNNs), modern solutions are based on semantic segmentation, providing pixel-level classification. Our solution, applicable also to other segmentation problems different from oil spill detection, is based on an innovative method based on image-to-image translation using different CNNs trained with an adversarial loss function for improving the model classification capabilities.

2. PROPOSED SOLUTION

In order to detect oil spill in SAR images, which basically consists in a segmentation problem, we have explored an approach based on *image-to-image translation*, performed through popular CNN architectures taken from the literature, tailored for the specific task. The networks that proved to reach the best performance in our problem were the following:

1. U-Net [5] architecture (Figure 1), quite popular for medical imaging;
2. U-Net++ [6] (Figure 2), an improved version of the U-Net that adds more nodes in the skip layers and offers the possibility to perform deep supervision. In our case, we added a final convolution layer to get the final output.

The above models were trained both with "standard" and adversarial learning. We used a large dataset consisting of 839 Radarsat-2 and Sentinel-1 SAR images over the Mediterranean Sea and some areas of the Atlantic Ocean and the North Sea, from which 2543 tiles containing oil spills

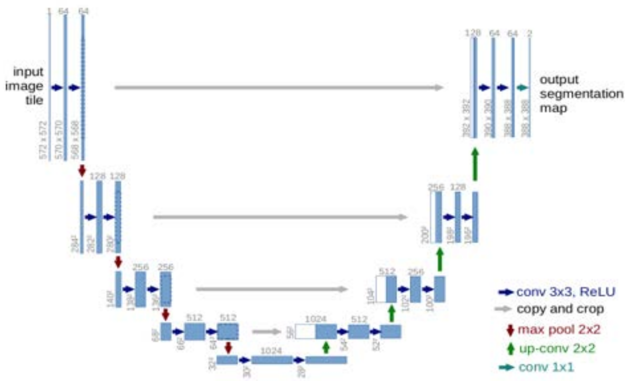

Figure 1: U-Net architecture. In the descending path, features are extracted, while in the ascending path image is reconstructed.

were extracted. Furthermore, we added 300 tiles containing only sea (including coast areas characterized by calm water, where it is easier to have false alarms). Each tile had a size of 256 x 256 pixels, and covered an area of at least 10 Km^2. The whole set of tiles was split for training and validation, according to the k-fold cross-validation approach, with k = 6.

When using "standard" learning, a combination of binary cross-entropy and dice loss was considered as loss function. For the adversarial learning approach, we implemented a Generative Adversarial Network (GAN) [4], which foresees the simultaneous training of two different models: a generator G performing the classification (the actual image to image translator), and a discriminative model D estimating the probability that a target is real or generated by the first net. The key idea is that each model tries to outperform the other one, participating in a minimaxing game.

In the literature, GANs were proved to be powerful, but adversarial training is slower than the standard one due to the presence of two networks, and convergence can be problematic, since if one of the nets (usually the discriminator) learns too fast and prevails, the other one will not improve. For these reasons, researchers and

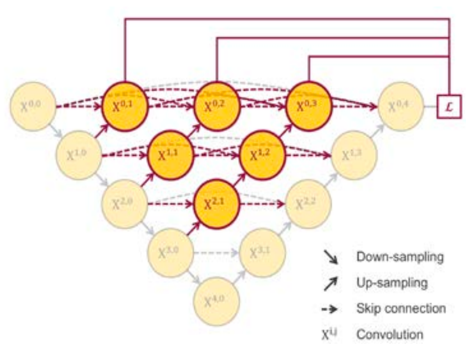

Figure 2: U-Net++ architecture: it can be seen as a "nested" U-Net. The upper nodes can be joined together for deep supervision.

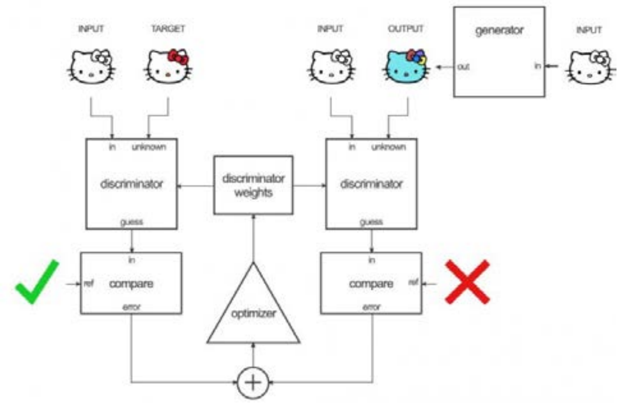

Figure 3: Pix2Pix working diagram. The discriminator observes a pair (image, ground truth mask) and a pair (image, generated mask) labeled as true or false. The weights are then propagated back to improve both generator and discriminator. Image taken from [7].

programmers often tend to avoid this solution, choosing a simpler solution. However, the use of GANs strongly improves image-to-image translation problems, and it was our idea that we could similarly improve also our segmentation task. Among the possible GANs, we chose the Pix2Pix model [7] (Figure 3), an architecture based on discriminating the generated mask in "patches" as real or fake: the discriminator D labels each patch corresponding to a receptive field of 70 × 70 pixels, and the result of each patch is averaged to make the final prediction. Original Pix2Pix implementation requires U-Net as its generator, but as aforementioned, we managed to use also its nested version, a solution never tried before.

3. RESULTS

The capability of the proposed method to discriminate between the two classes, oil spill and sea, was measured according to the Accuracy and Jaccard performance indices. The Accuracy index represents the percentage of pixels rightly classified as oil spill and water, while the Jaccard index (also known as Intersection over Union) is the ratio between the areas of the intersection and of the union, respectively, between the predicted mask and the ground truth. The Jaccard index is one of the most commonly used metrics in semantic segmentation, since it does not suffer from class imbalance.

As already mentioned, the set of ground truths used for validation was obtained, according to the k-fold cross-validation approach, with k = 6, from the set of tiles described in the previous section. However, we calculated the Accuracy and Jaccard indices only on a subset of the ground truths resulting from the cross-validation scheme. In fact, the set of ground truths includes two types of masks: either manually drawn by photointerpreters, or obtained using a semi-automatic segmentation tool. The manually drawn masks do not follow accurately the border of the spill,

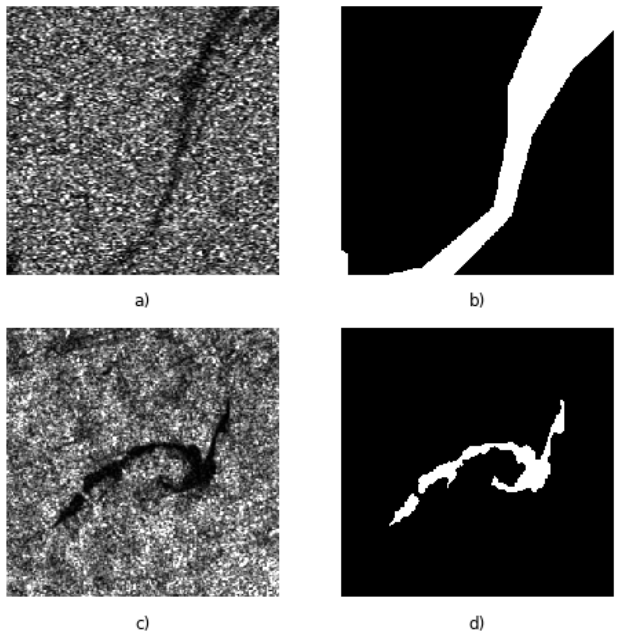

Figure 4: Examples of oil spills (a) and (c) and the corresponding masks generated manually (b) or automatically (d).

whereas the masks generated by the automatic segmentation tool are more accurate (see Figure 4). Both kinds of masks were used for training, because adding some "noise" allowed regularizing the semantic segmentation tasks. On the contrary, for validation we used only the automatically generated masks (about 72% of the total), in order to obtain a more accurate estimation of the Accuracy and Jaccard indices.

The validation results are reported in Table 1. In particular, in the column labeled "spill labeled", the index for the pixels labeled as spill is considered, while in the column "mean Jaccard" the average of the Jaccard indices for oil spill and sea classes is reported. It can be noted that the U-Net++ architecture performed better than its standard version, but the best results were obtained when training the standard U-Net model by adversarial learning. The results obtained by our approach outperformed previous solutions we found in the literature, which were based on SVM [2] and CNN [8][9] techniques (without adversarial training). More precisely, in [2] and [8], the accuracy index (the only reported) was around 94%, against our accuracy index of 98.3%. In [9] the authors got the best result using a simple U-Net architecture, obtaining a Jaccard index (the only reported) of 53.79, while our solution reaches a Jaccard value almost 20 points greater. Examples of segmentation performed by U-Net trained with adversarial learning are shown in Figure 5. To be noted that the use of adversarial learning was not as successful for the U-Net++ architecture as it was for the simpler U-Net. Further analysis has to be performed to understand better if adversarial learning can be applied also on the more complex U- Net++ model.

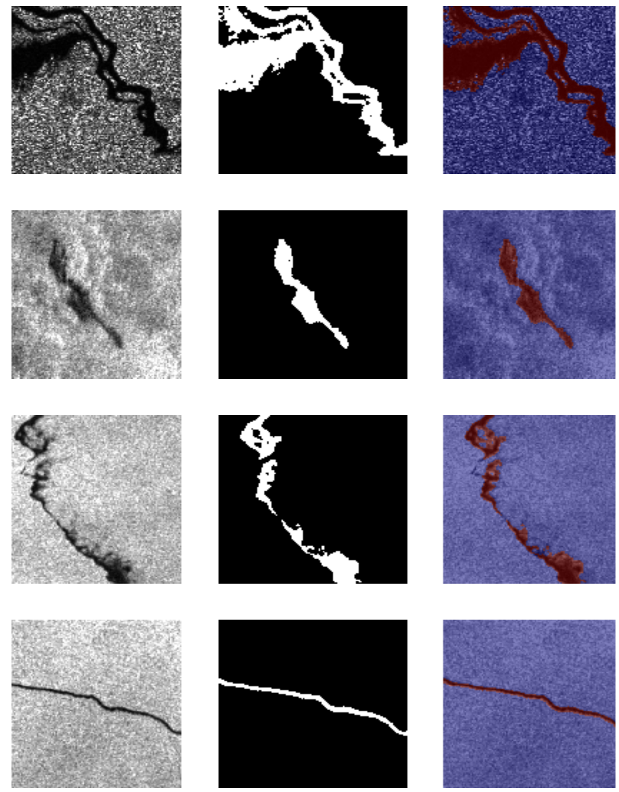

Figure 5: Examples of segmentation using the U-Net trained via adversarial network (GAN). On the left column, the oil spill as it appears in the SAR image, in the middle column the segmentation performed by our model, and on the right the segmentation overlapped with the original image

Architecture	**Spill labeled Jaccard index**	**Mean Jaccard index**	**Accuracy index**
U-Net	0.592	0.775	0.975
U-Net ++	0.689	0.817	0.983
U-Net (GAN)	0.729	0.820	0.983
U-Net++ (GAN)	0.469	0.702	0.973

Table 1: Table with test result on automatically-generated masks.

4. CONCLUSIONS

A new semantic model solution has been proposed with a successful application to the task oil spill detection. Using a modified version of the generator in the GAN architecture led to an improvement, in terms of visual results, accuracy and Jaccard index scores, compared to state of the art solutions. Improvements are still possible, for example through a more accurate post-processing that is subject of the current work.

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